

tached to it. Just place the soldering iron into the holder, pull the iron, and insert a new tip. It reduces the idle time and increases operator safety.

Weller also offers something similar. Its WE 1010 comes with a plastic knurled nut located on the soldering pencil by hand. Operator can turn the nut to change the tip. This process also increases the efficiency and reduces time for completing a job.

Soldering stations enable total control over the soldering iron and the working temperature. Some advanced soldering stations also offer calibration options for temperature control and smart systems that can work as a guide as well.

For example, JBC offers a soldering station with an easy-to-use menu, which displays tools preset independently for each tool type and port. The system analyses the soldering process to help the operator select the best tip for work and see the quality of each soldering joint.

Traceability

Soldering stations are being configured to manage production as well. The traceability feature introduced by Kurtz Ersä and JBC monitors work with power and temperature data in real time.

The i-CON TRACE by Kurtz-ersä is a fully networkable soldering station with an integrated WLAN, Bluetooth, and network card. It can be fully integrated into MES-controlled production processes, making the complete hand soldering process traceable and documentable.

i-CON TRACE can be connected



Fig. 4: Intuitive and safe operation of the i-CON TRACE with the Ersa TRACE app via WLAN. Setting and displaying all parameters on any number of soldering stations. For iOS and Android. In real time. With password protection (Credit: Kurtz Ersä)

to a manufacturing execution system (MES) with individually configured settings. This means seamless integration of the i-CON TRACE into networked production processes, which are already controlled by an MES.

Similarly, JBC offers a Web Manager feature, which allows firms to manage JBC soldering stations. This allows you to access station data remotely, using the company network and/or internet from any device, smartphone, tablet, Android, iPhone/iPad, or a computer.

Digital technology

Incorporating digital technology into the soldering iron is of huge interest as well. Digital soldering irons are becoming increasingly popular due to their accuracy and precision. Temperature calibration, for example, is one feature embedded in digital irons. The device digitally calibrates the difference between the set temperature and the tip temperature as measured by

Cordless Irons

Cordless irons are emerging onto the market as they are extremely portable and great for hard-to-reach spots. But they do not have unlimited run time. And their batteries need to be changed over time.

the tip thermometer.

These soldering irons come equipped with an LCD screen that displays the temperature and other vital information about the soldering process. The digital temperature control system ensures that the soldering iron maintains a consistent temperature throughout the job.

Digital soldering irons are also equipped with advanced safety features, such as automatic shut-off systems, that prevent the soldering iron from overheating and causing damage.

Finally, soldering irons are becoming more specialised to meet specific industry needs. For example, there are now soldering irons designed specifically for jewelry making, circuit board repair, and automotive work. These specialised soldering irons come with features that are tailored to the specific tasks they are designed for, making them more efficient and effective. **EFY**

This article has been put together by Jay Soni, an electronics enthusiast at EFY

Cleaning of Soldering Irons

Cleaning soldering irons is an important part of the complete work-rework process as this can affect the quality of tip and the developing device as well. The soldering kit generally comes with a simple sponge that cleanses the iron, but that is not quite efficient.

Shoritsu Electric Industry, or Senpujin, has designed a cyclone type soldering iron cleaner. A vortex-like air flow is generated inside the cleaner and dirt on the soldering iron tip is removed without contact. It also requires minimal maintenance.